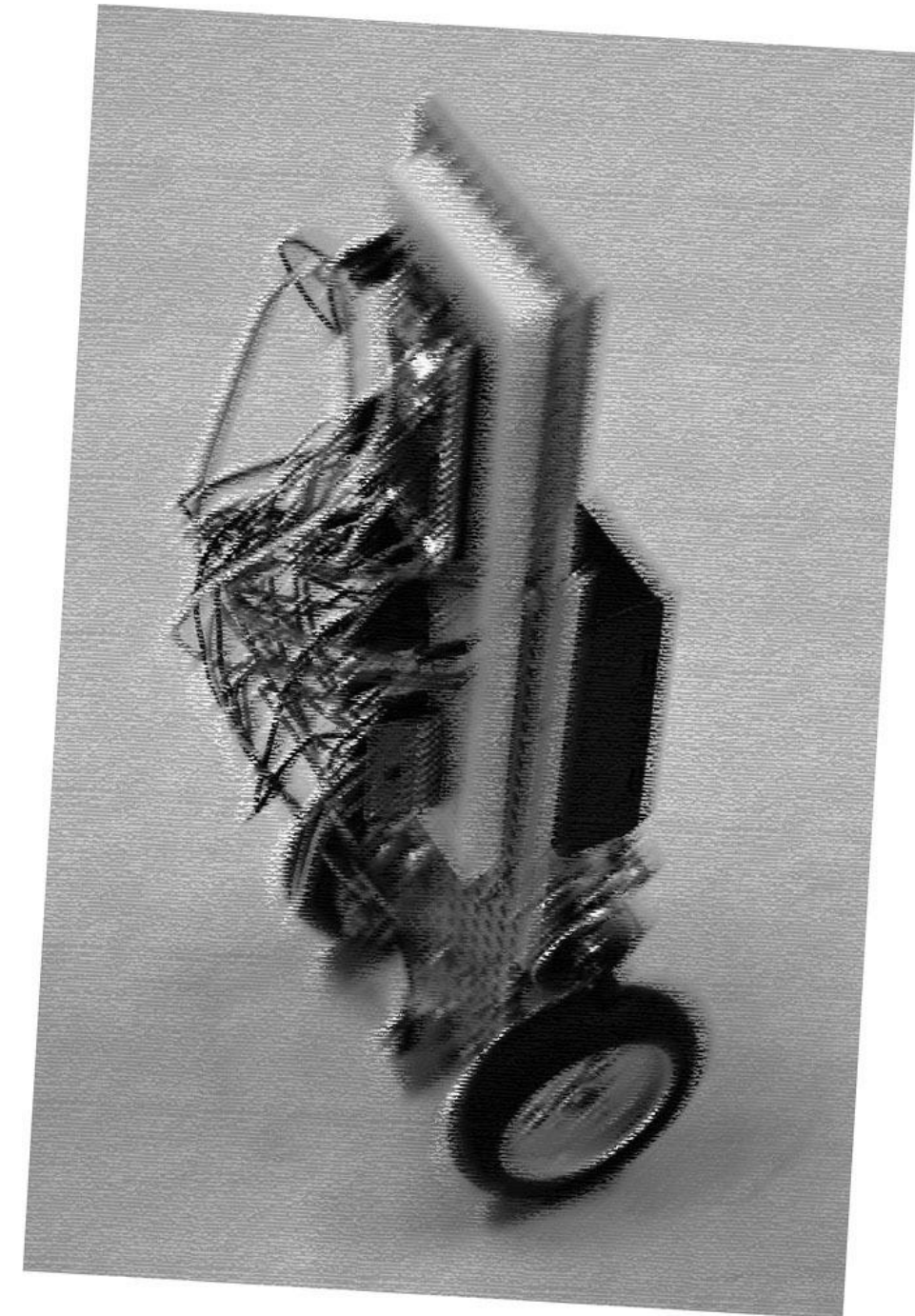


PRACTICAL EXERCISE · WEEK 2

Construction of a Wheeled Inverted Pendulum

Shaoqi (シャオ チー)

COURSE Department of Precision Engineering · 2026 S1S2
SESSIONS Mondays, 13:00–18:00 · Room 326, Engineering Bldg. 14
TODAY Course Introduction — 10 minutes



THE FOUR-WEEK ARC

Timeline

1

Week 1 Standing

Assembly & bring-up
Angle-only **standing** demo

2

Week 2 Position control

Build a rotary encoder
Position control
Simulation in Simulink

3

Week 3 EXTENSION

Load & disturbance tests
Advanced control (LQR/MPC)
.....

4

Week 4 presentation

10-minute presentation
Live demo & Q&A
Final grade · 40%

Week 3 is for extended tasks, or you can continue working on the tasks of Week 1 and Week2

Week 2

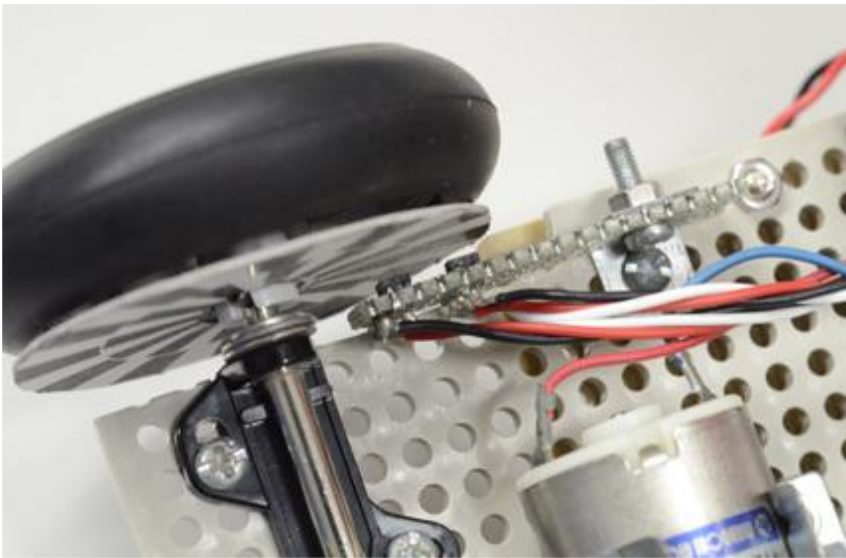
Assign the tasks among yourselves.

A

TASK A

STUDENT A · Hardware

Encoder Fabrication

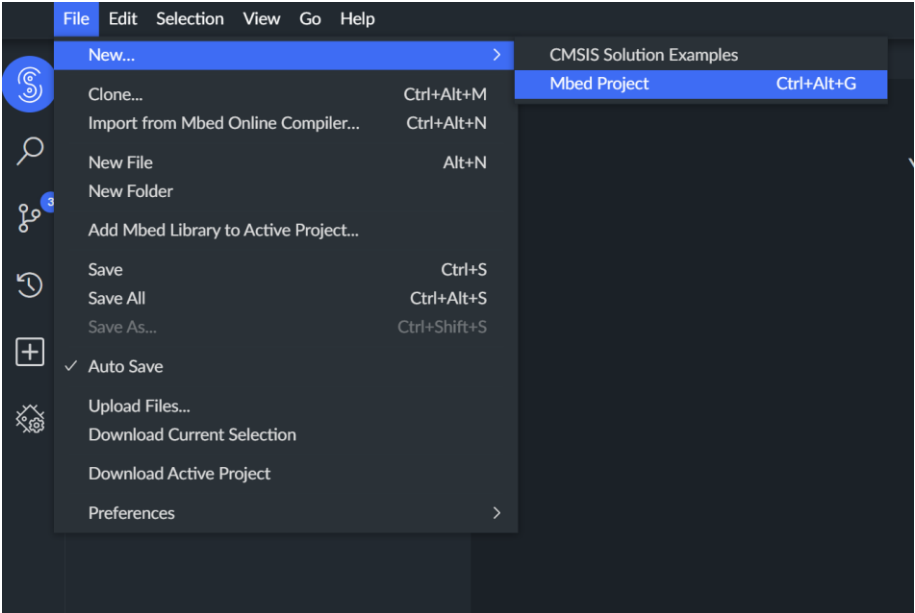


B

TASK B

STUDENT B · FIRMWARE

Add Encoder Feedback to
Microcontroller

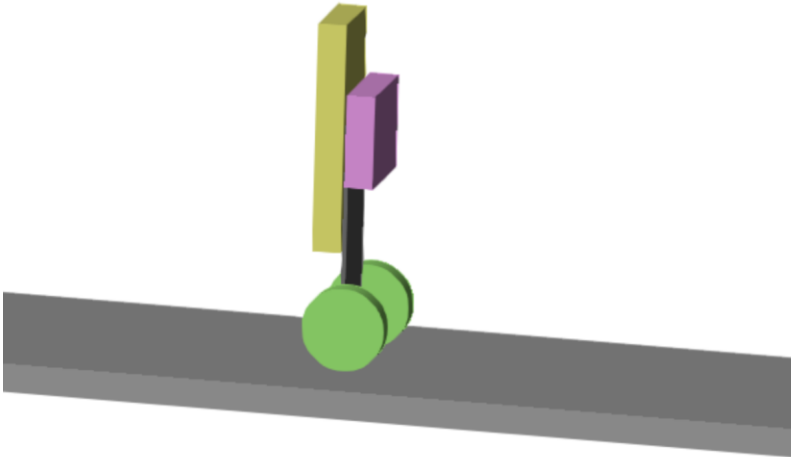


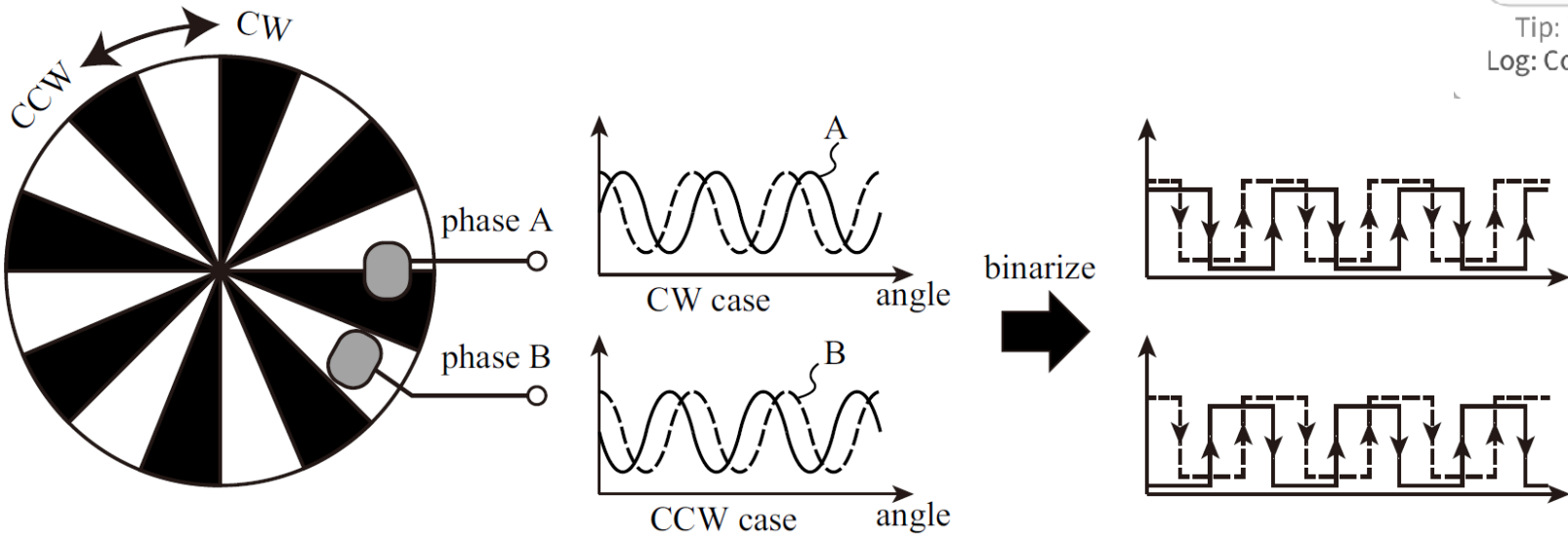
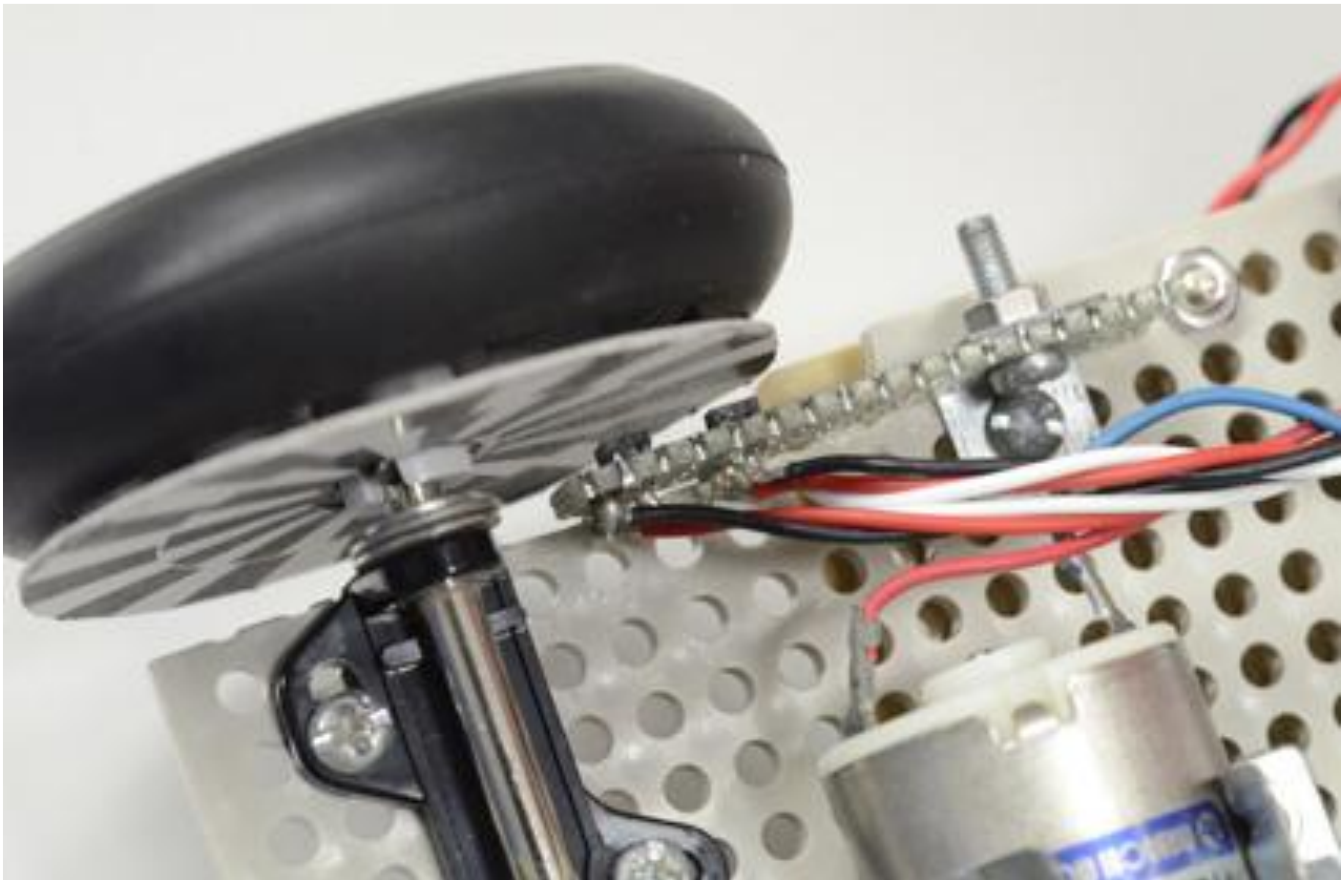
C

TASK C

STUDENT C · SIMULATION

Simulink Simulation





analog_input_visualizer

4ch Analog Debug (Frame: AA 55 CH1 CH2 CH3 CH4)

Port: COM3 (click to cycle)

Refresh

Disconnect

Decoded Channels (raw 0..255) [CH1=p16, CH2=p17, CH3=p19, CH4=p20]

CH1 (p16): 79	CH2 (p17): 62	CH3 (p19): 151	CH4 (p20): 10
CH1: 0.310	CH2: 0.243	CH3: 0.592	CH4: 0.039

Frames: 897 Bad bytes: 0 (auto re-sync by header)

Simple Bar View (0..1)

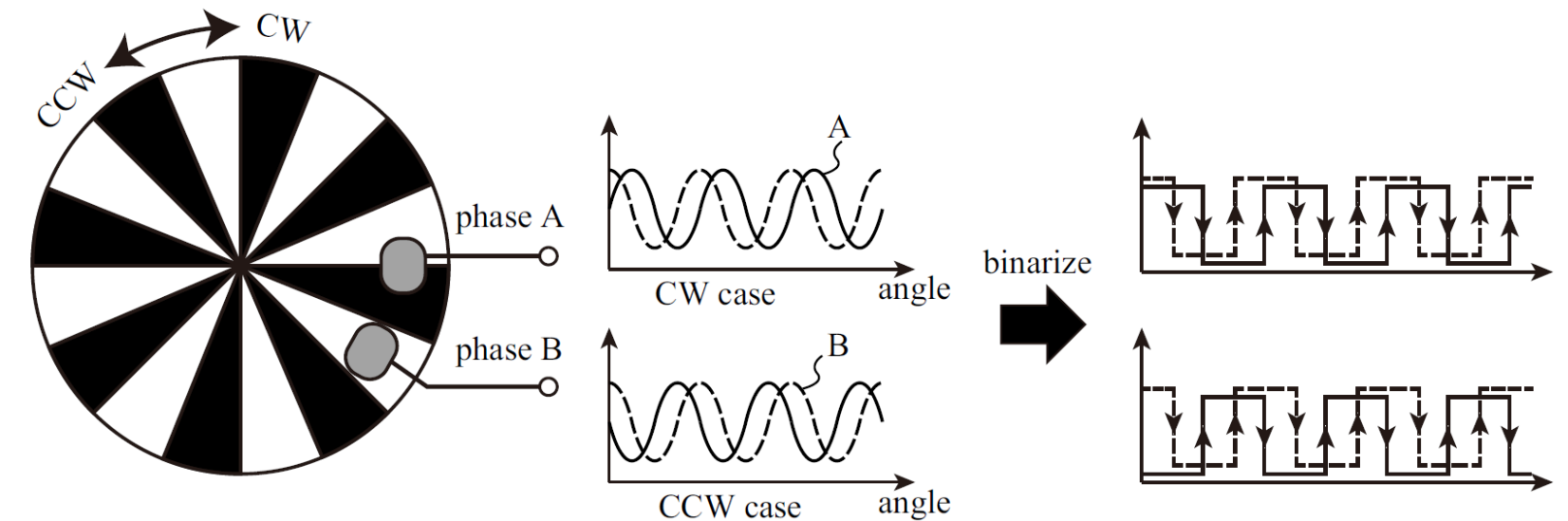
CH1 p16		0.310
CH2 p17		0.243
CH3 p19		0.592
CH4 p20		0.039

Tip: if values look frozen -> check baud / wiring / device is sending frames.
Log: Connected: COM3 @ 115200


```

3 // Sampling frequency
4 #define SAMP_FREQ 2000.0f
5 // Motor voltage limit ( $V_{\text{motor}} = 3.3 * v_{\text{out}} * 4$ )
6 #define VLIMIT 0.1f
7 // Motor offset voltage
8 #define VOFFSET 0.0f
9 // Feedback gains (Arbitrary Unit)
10 #define KP 1.0f
11 #define KD 0.0f
12

```

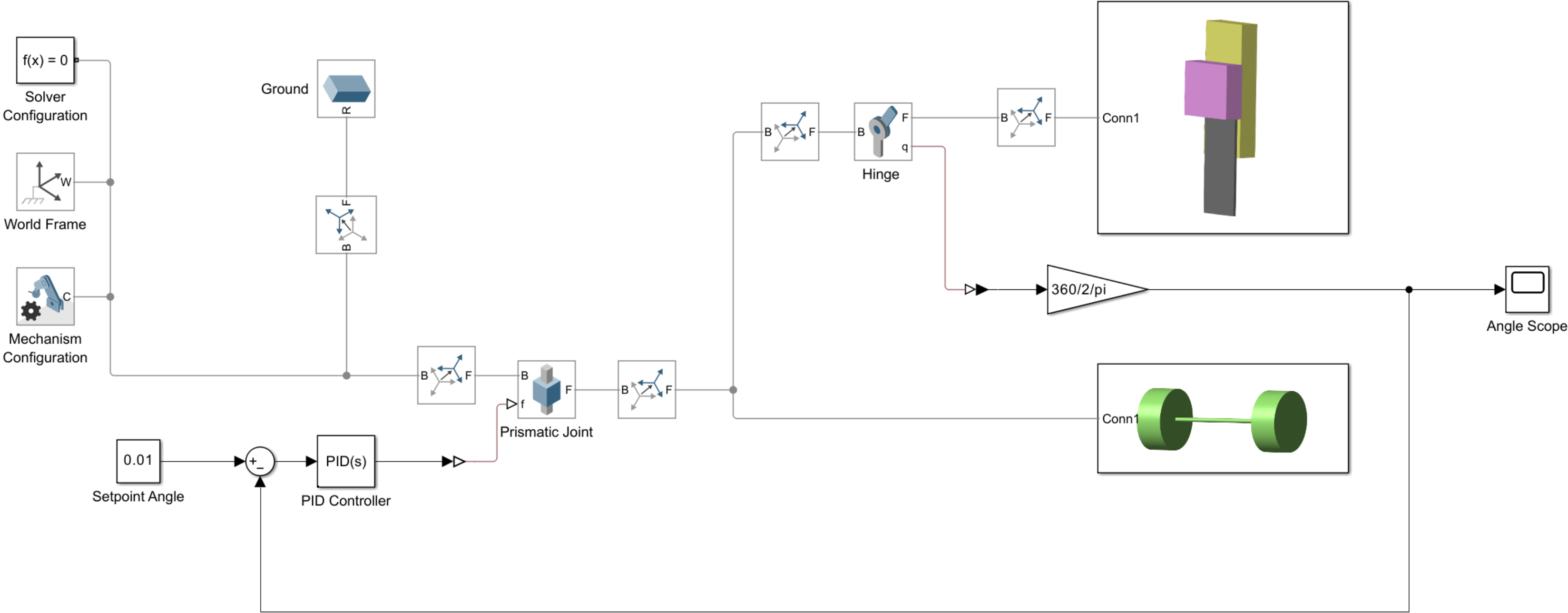


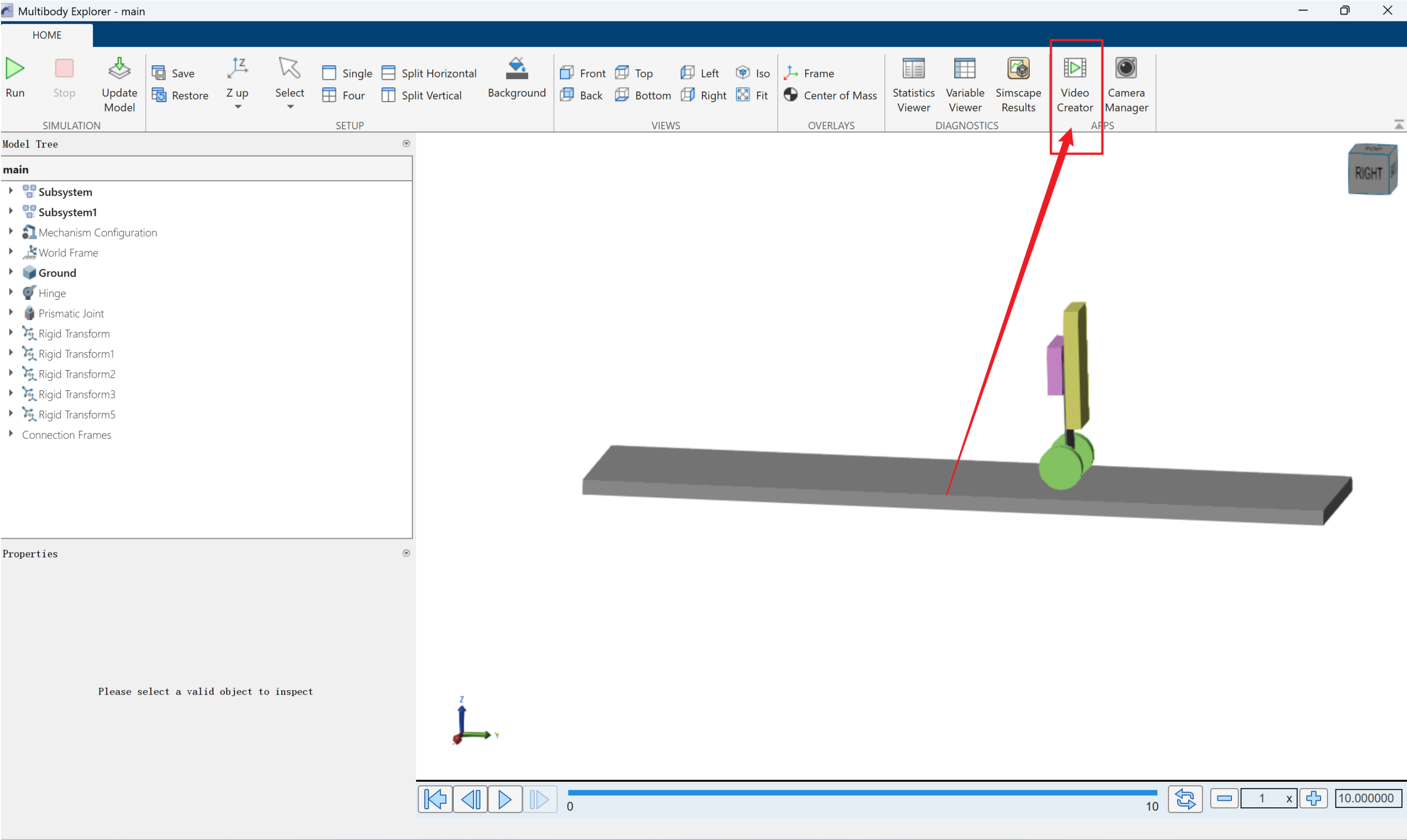
```

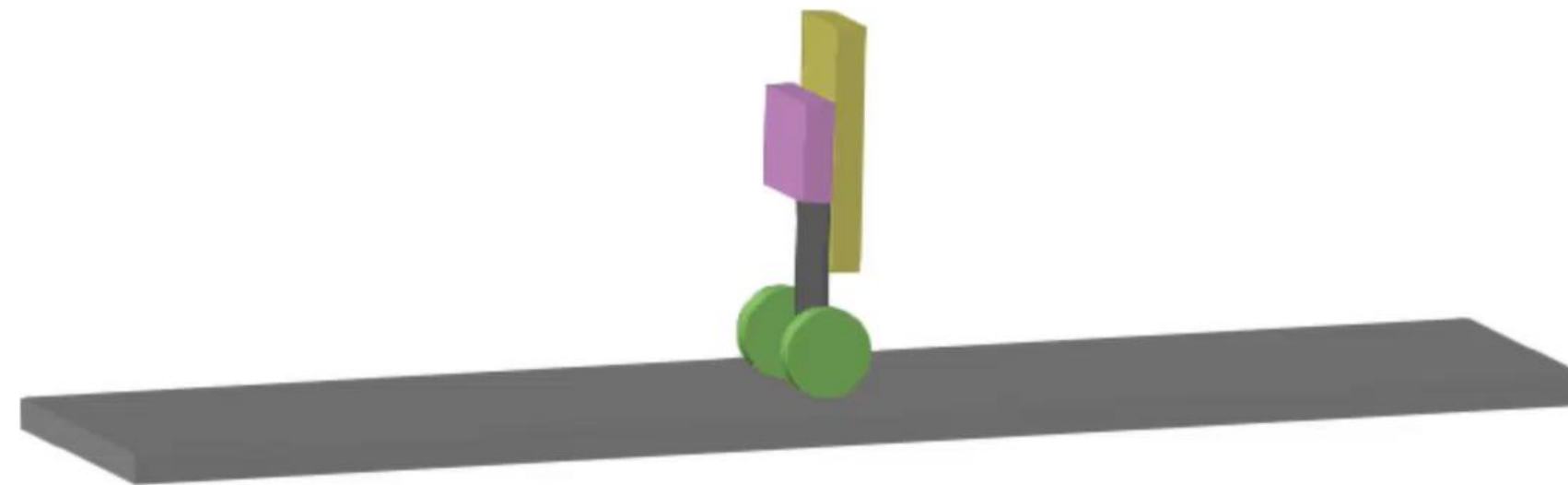
105 // Motor output
106 if (pd1v < 0.02f && pd2v < 0.02f) { // if not standing, stop the motor
107     motor_v = 0;
108     led_v_forward = 1;
109     led_v_backward = 1;
110 } else {
111     motor_v = v_out + VOFFSET;
112     led_v_forward = (v_dir > 0 ? v_out * 3.3f : 0.0f);
113     led_v_backward = (v_dir < 0 ? v_out * 3.3f : 0.0f);
114 }
115

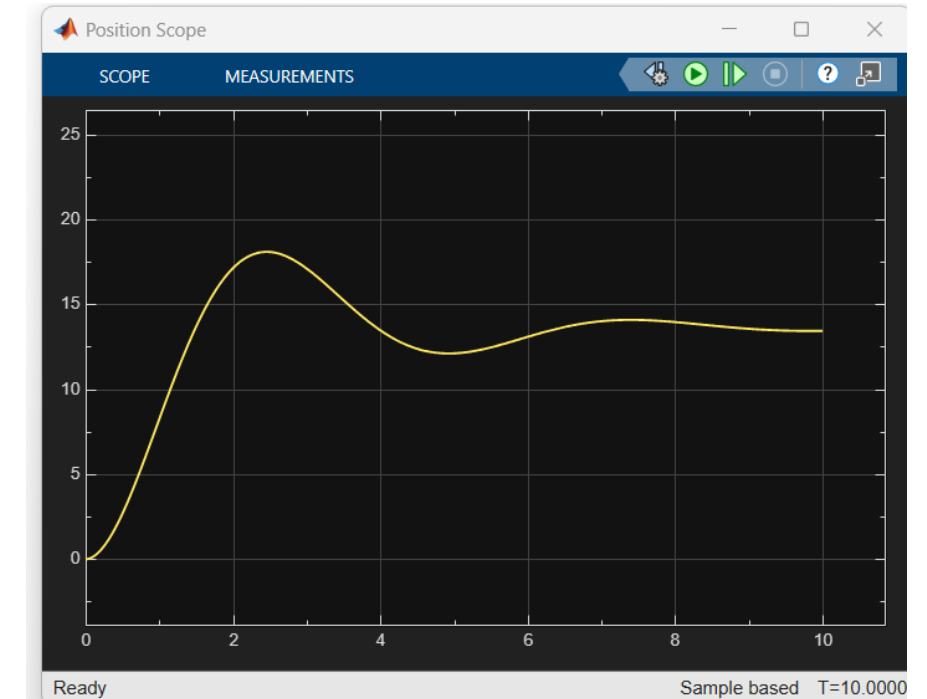
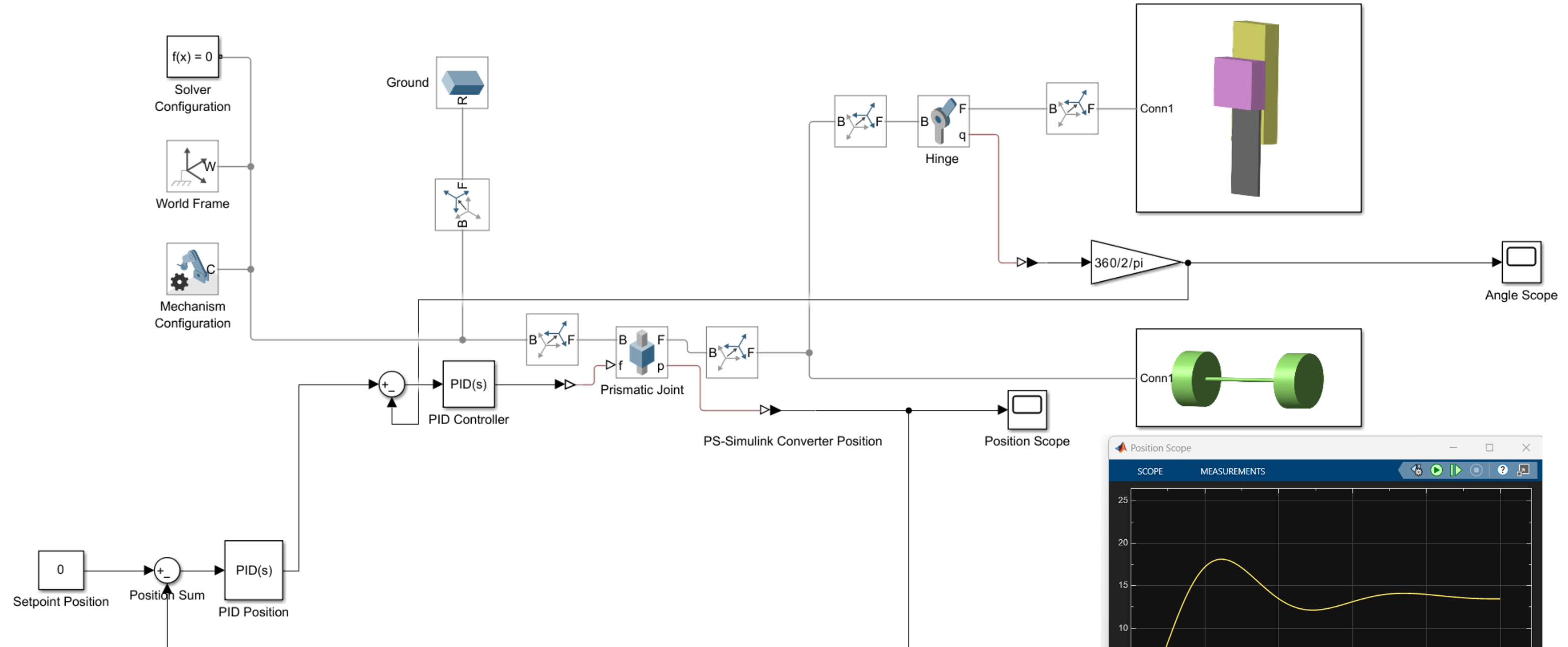
```

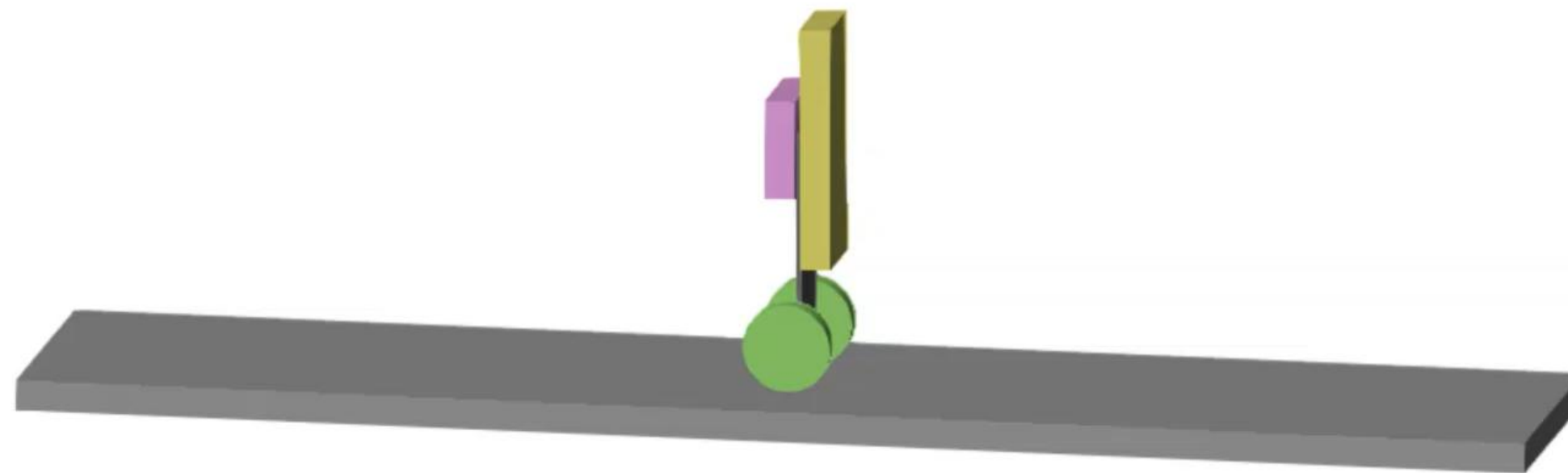
Use AI Tools to help you understanding the codes



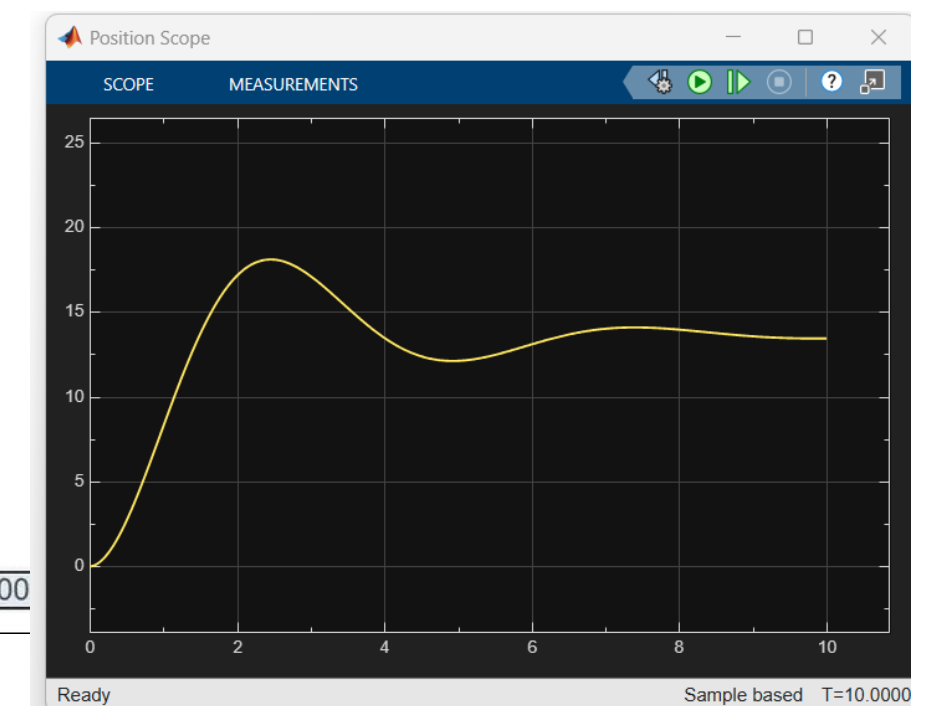








0.0000



Week 2 Simulink Simulation – Disturbance

11

Block Parameters: Hinge Disturbance Torque

Pulse Generator

Output pulses:

```
if (t >= PhaseDelay) && Pulse is on
    Y(t) = Amplitude
else
    Y(t) = 0
end
```

Pulse type determines the computational technique used.

Time-based is recommended for use with a variable step solver, while Sample-based is recommended for use with a fixed step solver or within a discrete portion of a model using a variable step solver.

Parameters

Pulse type: Time based

Time (t): Use simulation time

Amplitude: 0.05

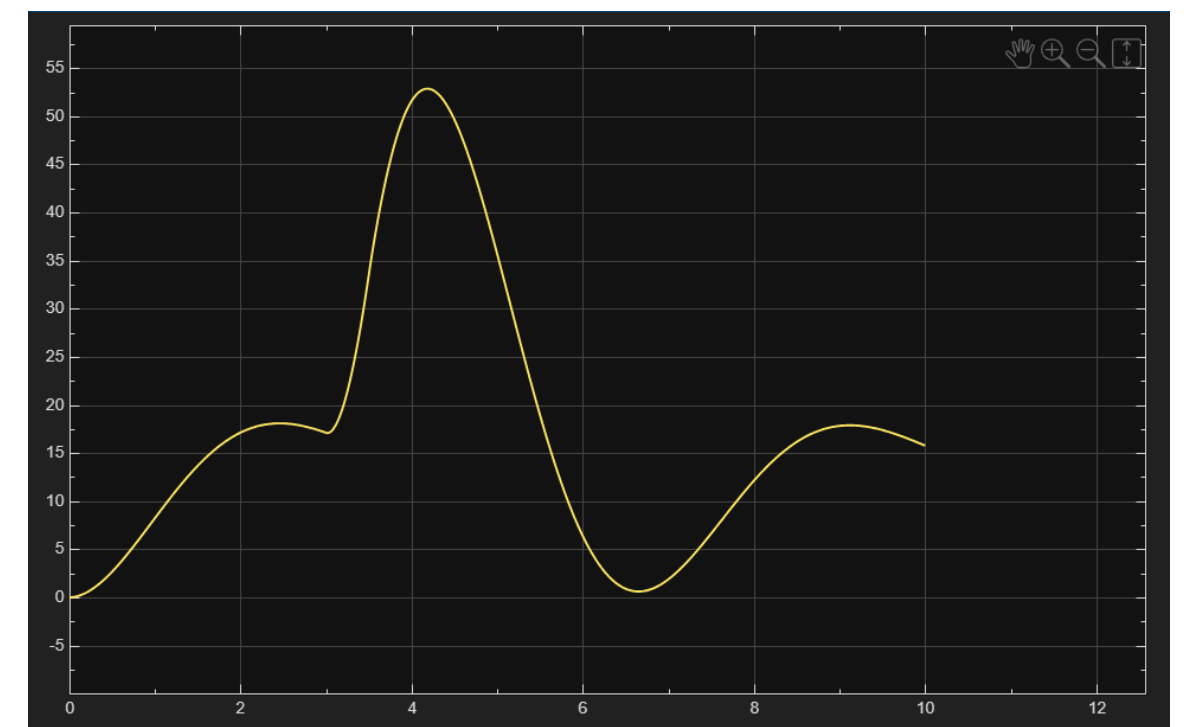
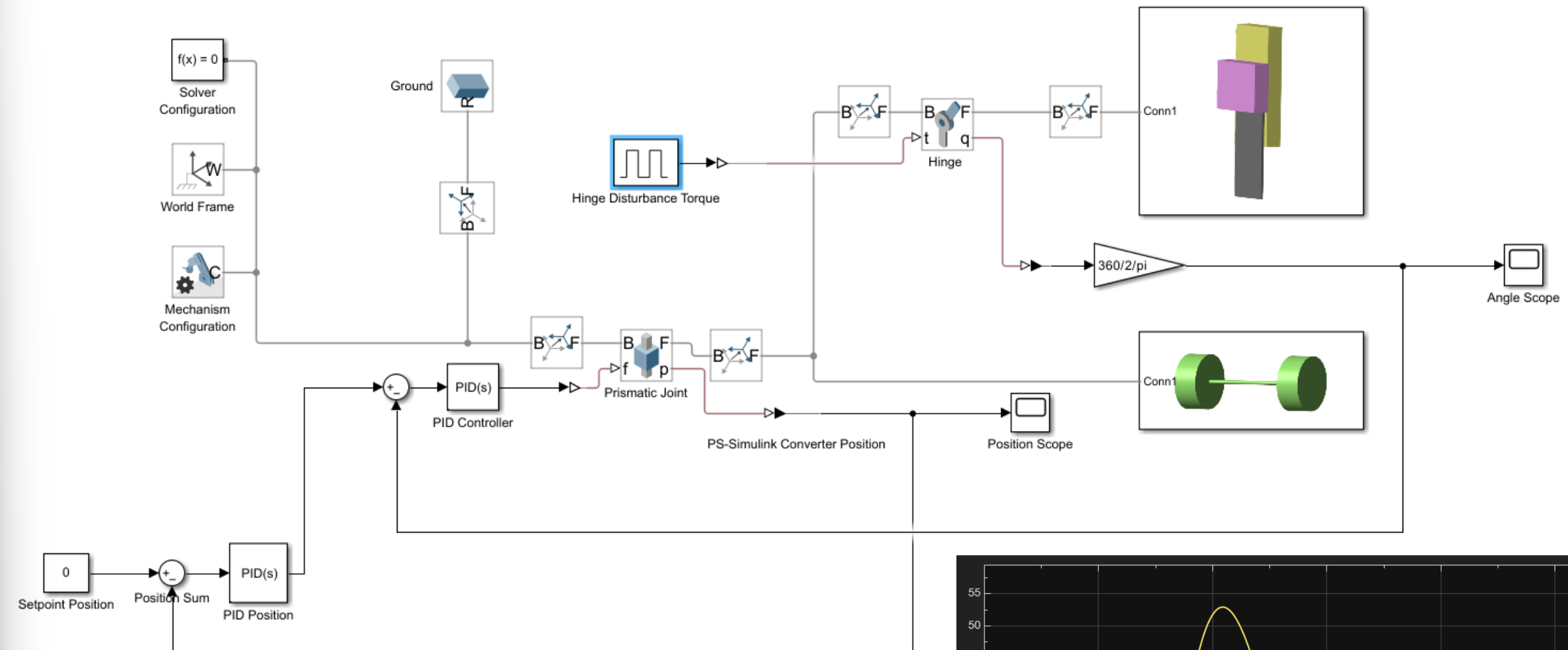
Period (secs): 10

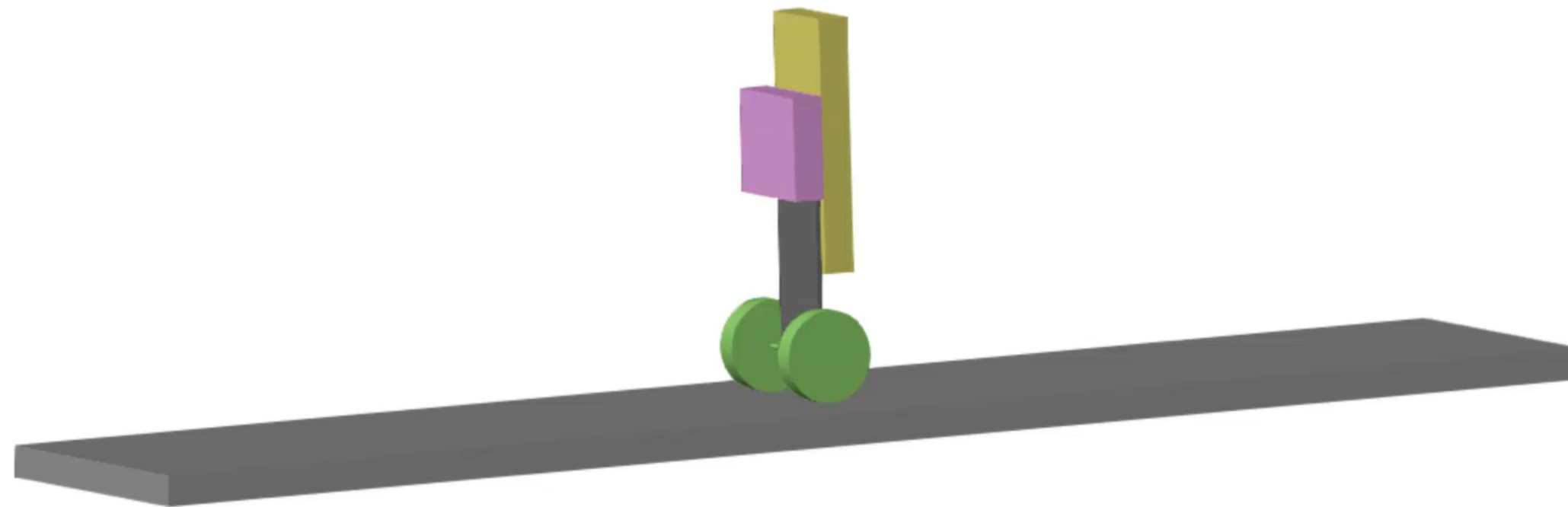
Pulse Width (% of period): 5

Phase delay (secs): 3

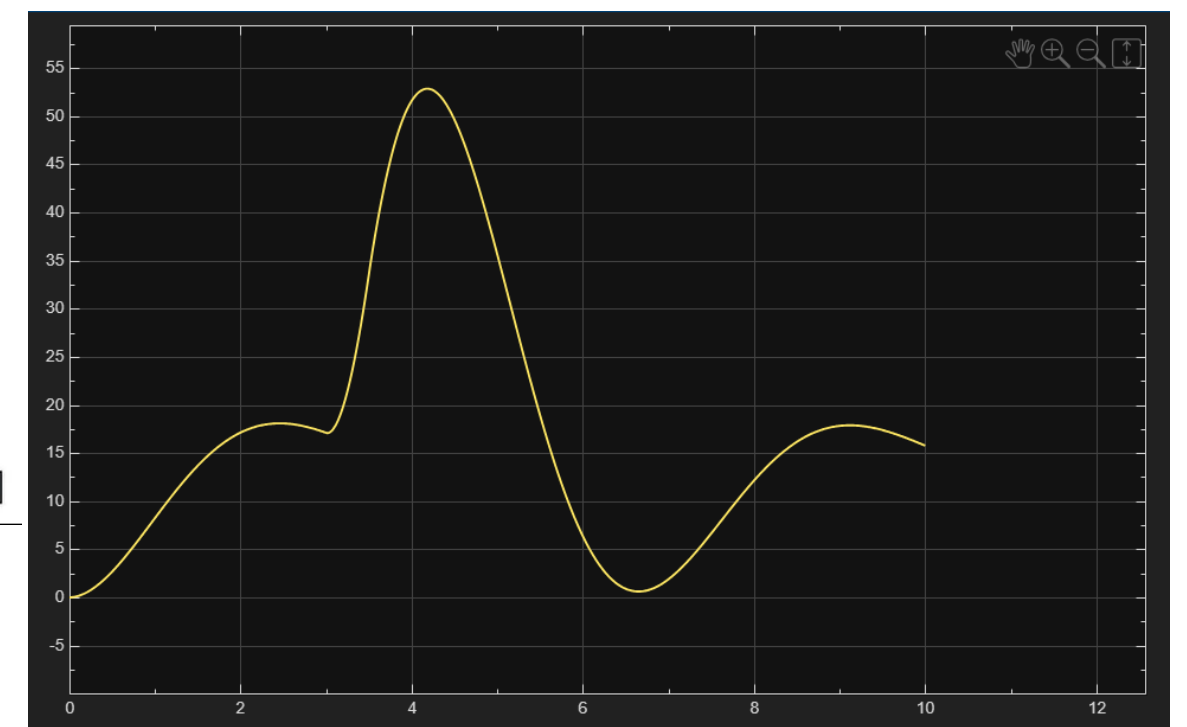
☒ Interpret vector parameters as 1-D

OK Cancel Help Apply

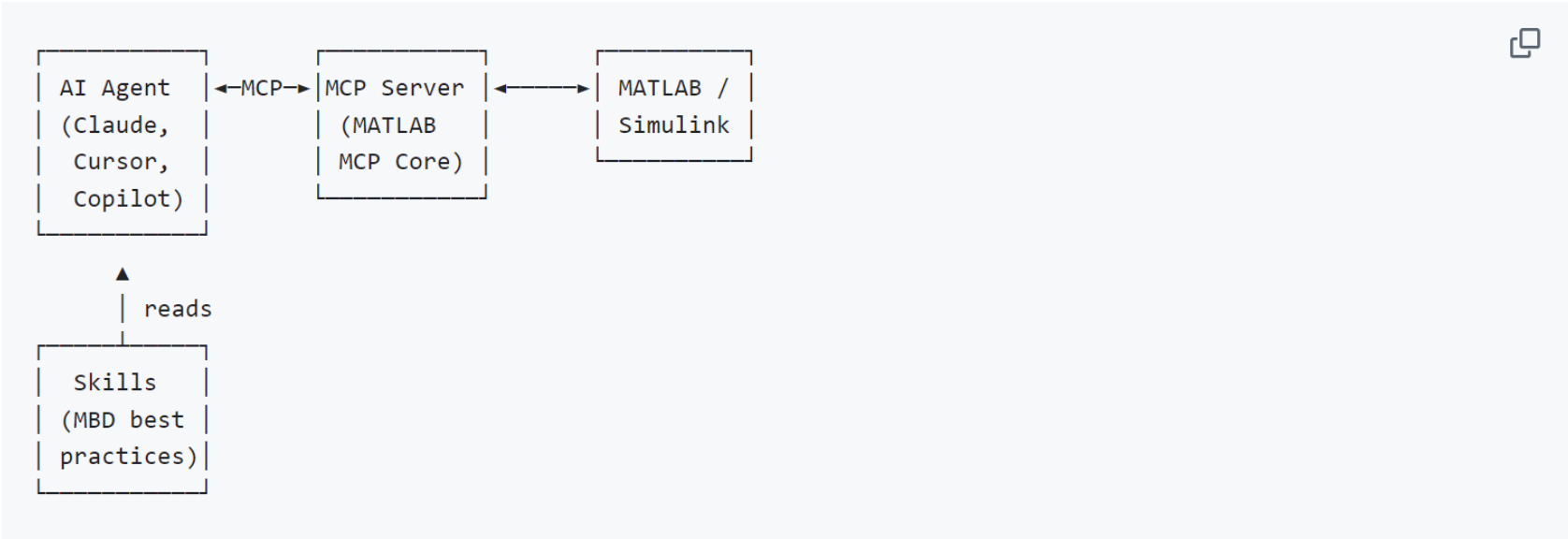




0.000000



How It Works



Your agent reads **skills** for domain knowledge, then calls **MCP tools** to interact with MATLAB and Simulink. The [MATLAB MCP Core Server](#) bridges the connection (downloaded during setup).

Supported Platforms

Platform	Setup	Notes
Claude Code	Automated	Also supports no-clone marketplace install (skills only)
GitHub Copilot	Automated	
OpenAI Codex	Automated	
Gemini CLI	Automated	
Sourcegraph Amp	Automated	
Cursor	Manual	Untested

<https://github.com/matlab/simulink-agentic-toolkit>

Command Window

```
matlab.addons.install('C:\Users\rocke\.local\share\MATLABMCPCoreServerToolkit.mltbx')
addpath('D:\Local\20251230_Inverted_Pendulum\Week2\simulink-agentic-toolkit')
satk_initialize

ans =

1x4 table

      Name      Version  Enabled  Identifier
      _____  _____  _____  _____
"MATLAB MCP Core Server Toolbox"  "0.1.0"  true  "fce99a56-1b63-4d49-9d70-c32c5605e4eb"
```

Simulink Agentic Toolkit - Installation Check
=====

Prerequisites

- ✓ MATLAB 25.2.0.3055257 (R2025b) Update 2
- ✓ Simulink installed

MATLAB Environment

- ✓ All 6 tool entry points on path
- ✓ Package resolution correct (+sage, +mbd inside SATK)

MCP Server

- ✓ matlab-mcp-core-server.exe found: C:\Users\rocke\.local\bin\matlab-mcp-core-server.exe

MCP Connectivity

- ✓ shareMATLABSession available (MATLABMCPCoreServerToolkit installed)
- ✓ MATLAB connector running (port 31516)

=====

Result: PASS

Week 2 Submission

Video 1:

show the real wheeled inverted pendulum cart under a disturbance.

Video 2:

show the simulation running under normal conditions without an external disturbance.

Video 3:

show the simulation response when a disturbance is applied.

DEADLINE

21:00

on the day of the class
Today — May 7

Late submissions accepted until **21:00 the following day** . Late score capped at 50%.

Each group only needs one submission.